JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON THURSDAY 02nd APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

CHEMISTRY — TEST PAPER WITH SOLUTION

SECTION-A

51. The ratio of mass percentage (w/w) of C : H in a hydrocarbon is 12 : 1. It has two carbon atoms. The weight (in g) of $CO_2(g)$ formed when 3.38 g of this hydrocarbon is completely burnt in oxygen is :

(Given : Molar mass in g mol^{-1} C : 12, H : 1, O : 16)

(1) 5.68 (2) 11.44

(3) 22.74 (4) 17.05

Sol. Molecular formula of hydrocarbon is C_2H_2

$$\text{Moles of } C_2H_2 = \frac{3.38}{26}$$

$$C_2H_2(g) + \frac{5}{2}O_2(g) \rightarrow 2CO_2(g) + H_2O(l)$$

$$\text{Moles of } CO_2 \text{ produced} = \left(\frac{3.38}{26} \times 2\right)$$

$$\text{Mass of } CO_2 \text{ produced} = \left(\frac{3.38}{26} \times 2\right) \times 44 = 11.44$$

52. The first and second ionization constants of a weak dibasic acid H_2A are 8.1×10^{-8} and 1.0×10^{-13} respectively. 0.1 mol of H_2A was dissolved in 1L of 0.1 M HCl solution. The concentration of HA^- in the resultant solution is :

(1) 0.1 M (2) 9.53×10^{-6} M

(3) 8.1×10^{-8} M (4) 1.0×10^{-13} M

Sol.

$$\underset{0.1M}{HCl} \rightarrow \underset{0.1M}{H^+} + \underset{0.1M}{Cl^-}$$

$$\underset{0.1-x}{H_2A} \rightleftharpoons \underset{x+0.1}{H^+} + \underset{x}{HA^-}$$

(Due to common ion effect x is very less)

$$Ka_1 = \frac{[H^+][HA^-]}{[H_2A]}$$

$$Ka_1 = \frac{0.1 \times [HA^-]}{0.1}$$

$$[HA^-] = Ka_1 = 8.1 \times 10^{-8}$$

53. SF_4 is isostructural with :

A. $BrF_4^{\ominus}$

B. CH_4

C. $IF_4^{\oplus}$

D. XeF_4

E. XeO_2F_2

Choose the **correct** answer from the options given below :

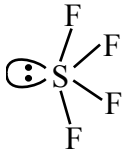
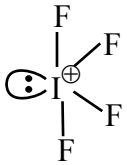
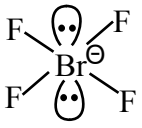
(1) C only

(2) C and E only

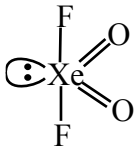
(3) A and D only

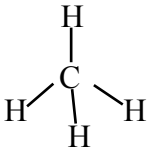
(4) B and E only

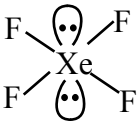
Sol.

Species	Structure / Shape
SF_4	 → see-saw
$IF_4^{\oplus}$	 → see-saw
$BrF_4^{\ominus}$	 → square planar

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XeO_2F_2  → see-saw

CH_4  → Tetrahedral

XeF_4  → square planar

54. Gas 'A' undergoes change from state 'X' to state 'Y'. In this process, the heat absorbed and work done by the gas is 10 J and 18 J respectively. Now gas is brought back to state 'X' by another process during which 6 J of heat is evolved. In the reverse process of 'Y' to 'X',

(1) 18 J of the work is done by the gas 'A'.

(2) 2 J of the work is done by the gas 'A'.

(3) 12 J of the work is done on the gas 'A' by the surrounding.

(4) 14 J of the work is done on the gas 'A' by the surrounding.

Sol. For process X → Y

$w = -18$ J

$q = +10$ J

$\Delta U = q + w$

$\Delta U_{X\to Y} = 10 + (-18) = -8$ J

For reverse process :

$\Delta U_{Y\to X} = 8$ J

$q = -6$ J

$\Delta U = q + w$

$8 = -6 + w$

$w = 14$ J

55. Solution A is prepared by dissolving 1 g of a protein (molar mass = 50000 g mol^{-1}) in 0.5 L of water at 300 K. Its osmotic pressure is x bar. Solution B is made by dissolving 2 g of same protein in 1 L of water at 300 K. Osmotic pressure of solution B is y bar. Entire solution of A is mixed with entire solution of B at same temperature. The osmotic pressure of resultant solution is z bar. x, y and z respectively are :

($R = 0.083$ L bar mol^{-1} K^{-1})

(1) 9.96×10^{-4}; 9.96×10^{-4}; 9.96×10^{-4}

(2) 9.96×10^{-4}; 9.96×10^{-4}; 19.92×10^{-4}

(3) 4.98×10^{-4}; 4.98×10^{-4}; 9.96×10^{-4}

(4) 4.98×10^{-4}; 4.98×10^{-4}; 4.98×10^{-4}

Sol. $'x' = \dfrac{1}{50,000 \times 0.5} \times R \times 300 = 9.96 \times 10^{-4}$ bar

$'y' = \dfrac{2}{50,000 \times 1} \times R \times 300 = 9.96 \times 10^{-4}$ bar

$'z' = \dfrac{'x' \times \frac{1}{2} + 'y' \times 1}{\frac{3}{2}} = 9.96 \times 10^{-4}$ bar

56. At 25°C, 20.0 mL of 0.2 M weak monoprotic acid HX is titrated against 0.2 M NaOH. The pH of the solution (a) at the start of the titration (when NaOH has not been added) and (b) when 10 mL of NaOH is added respectively, are :

Given : $K_a = 5 \times 10^{-4}$, $pK_a = 3.3$, $\alpha << 1$

(1) 0.7, 2.0

(2) 2.0, 3.3

(3) 1.1, 2.2

(4) 3.0, 2.2

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Sol. For 0.2M, 20ml HX solution,

$$\text{pH of weak acid} = \frac{1}{2}[pk_a - \log c]$$

$$= \frac{1}{2}[4 - \log 5 - \log 0.2]$$

$$= \frac{1}{2}[4 - \log 5 + \log 5]$$

$$= 2$$

$$HX + \overset{(LR)}{NaOH} \longrightarrow NaA + H_2O$$

20 ml, 0.2 M		10 ml, 0.2 M
4m mole		2m mole
2m mole	–	2m mole

[HX + NaA] $\Rightarrow$ acidic buffer

$$pH = pK_a + \log \frac{salt}{acid}$$

$$pH = 3.3 + \log \frac{2}{2}$$

$$pH = 3.3$$

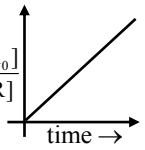
57. Consider the reaction aX $\rightarrow$ bY, for which the rate constant at 30°C is 1×10^{-3} mol^{-1} Ls^{-1}. Which of the following statements are true ?

A. When concentration of 'X' is increased to four times, the rate of reaction becomes 16 times.

B. The reaction is a second order reaction.

C. The half-life period is independent of the concentration of X.

D. Decomposition of N_2O_5 is an example of the above reaction.

E. $\ell n \frac{[R_0]}{[R]}$ 

is valid for the above reaction.

Choose the **correct** answer from the options given below :

(1) A and B only (2) A, B and C only

(3) A, B, D and E only (4) C and D only

Sol. Rate of constant (k) = 10^{-3} M^{-1} sec^{-1}

$\because$ 1 – n = –1 [unit of k = M^{1-n} sec^{-1}]

n = 2 $\Rightarrow$ 2nd order reaction

rate = $k[A]^2$

(A) If concentration of A is 4 times then reaction will become 16 times.

(B) Order of reaction is 2

(C) Half-life of the reaction is independent of the concentration of the reactant for 1st order, not for 2nd order reaction.

(D) Decomposition of N_2O_5 is example of 1st order reaction

(E) Graph of $\ell n \frac{[R]_0}{[R]_t}$ vs t is straight line for 1st order not for 2nd order reaction.

So statement (A) & (B) are correct

58. The correct set that contains all kinds (basic, acidic, amphoteric and neutral) of oxides is :

(1) Na_2O, K_2O, Al_2O_3 and As_2O_3

(2) Al_2O_3, As_2O_3, CO and NO

(3) K_2O, Cl_2O_7, As_2O_3 and NO

(4) Na_2O, N_2O, Al_2O_3 and CO

Sol. Basic Oxides : Na_2O , K_2O

Acidic Oxides : Cl_2O_7

Neutral Oxides : N_2O , NO & CO

Amphoteric Oxides : Al_2O_3 , As_2O_3

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59. Given below are two statements :

Statement-I : The second ionization enthalpy of B, Al and Ga is in the order of B > Al > Ga.

Statement-II : The correct order in terms of first ionization enthalpy is Si < Ge < Pb < Sn.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement-I** and **Statement-II** are true.

(2) Both **Statement-I** and **Statement-II** are false.

(3) **Statement-I** is true but **Statement-II** is false.

(4) **Statement-I** is false but **Statement-II** is true.

Sol. Order of II ionization energy for boron family elements is :

$$\underset{(2427\ KJ/mol)}{B} > \underset{(1979\ KJ/mol)}{Ga} > \underset{(1816\ KJ/mol)}{Al}$$

Order of I ionization energy for carbon family elements is :

$$\underset{(786\ KJ/mol)}{Si} > \underset{(761\ KJ/mol)}{Ge} > \underset{(715\ KJ/mol)}{Pb} > \underset{(708\ KJ/mol)}{Sn}$$

60. Given below are two statements :

Statement-I : Among Zn, Mn, Sc and Cu, the energy required to remove the third valence electron is highest for Zn and lowest for Sc.

Statement-II : The correct order of the following complexes in terms of CFSE is $[Co(H_2O)_6]^{2+} < [Co(H_2O)_6]^{3+} < [Co(en)_3]^{3+}$.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **Statement-I** and **Statement-II** are true.

(2) Both **Statement-I** and **Statement-II** are false.

(3) **Statement-I** is true but **Statement-II** is false.

(4) **Statement-I** is false but **Statement-II** is true.

Sol. **Statement I**

	IE_3 (kJ/mol)
Sc	2393
Mn	3260
Cu	3556
Zn	3837

Sc < Mn < Cu < Zn (3rd IE)

Statement II

CFSE $\Rightarrow$ SFL > WFL

(en > H_2O)

CFSE $\propto$ Charge on metal

$Co^{+2} < Co^{+3}$

$[Co(H_2O)_6]^{2+} < [Co(H_2O)_6]^{3+} < [Co(en)_3]^{3+}$

61. Which of the following complexes will show coordination isomerism ?

(A) $[Ag(NH_3)_2][Ag(CN)_2]$

(B) $[Co(NH_3)_6][Cr(CN)_6]$

(C) $[Co(NH_3)_6][Co(CN)_6]$

(D) $[Fe(NH_3)_6][Co(CN)_6]$

(E) $[Co(NH_3)_6][Fe(CN)_6]$

Choose the **correct** answer from the options given below :

(1) B, C and D Only

(2) B, D and E Only

(3) A, C and D Only

(4) C, D and E Only

Sol. Compound, $[Ag(NH_3)_2][Ag(CN)_2]$ cannot show coordination isomerism. All other complexes (B, C, D and E) will show co-ordination isomerism.

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62. Complete combustion of X g of an organic compound gave 0.25 g of CO_2 and 0.12 g of H_2O. If the % of carbon is 25% and of hydrogen is 4.89%, then X = _____ $\times 10^{-3}$ g. (Nearest integer)
(Molar mass of C, H and O are 12, 1 and 16 g mol^{-1} respectively.)
(1) 273 (2) 27
(3) 2730 (4) 227

Sol. Mass of carbon = $\frac{0.25}{44}\times 12$

Mass% of carbon = $\frac{\text{Mass of carbon}}{x}\times 100$

$x = \frac{\frac{0.25\times 12}{44}\times 100}{25}$

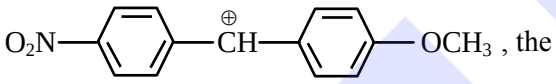
$x = \frac{12}{44}$

$x = 0.2727 = 272.7 \times 10^{-3}$

$\simeq 273 \times 10^{-3}$

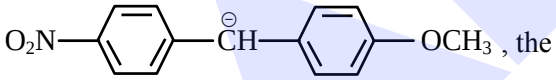
63. Given below are two statements :

Statement I : In



carbocation is stabilised by +R effect of $-OCH_3$ group.

Statement II : In

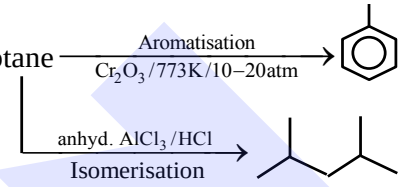


carbanion is stabilised by –R effect of $-NO_2$ group.
In the light of the above statements, choose the **correct** answer from the options given below :
(1) Both **Statement I** and **Statement II** are true
(2) Both **Statement I** and **Statement II** are false
(3) **Statement I** is true but **Statement II** is false
(4) **Statement I** is false but **Statement II** is true

Sol. Both statements are correct

64. The compound (X) on
(i) on heating in the presence of anhydrous $AlCl_3$ and HCl gas gives 2, 4–dimethyl pentane
(ii) aromatization gives toluene and
(iii) cyclisation gives methyl cyclohexane
The correct name of compound (X) is :
(1) Hept-2-ene (2) Hept-1,3,5-triene
(3) Heptane (4) Hept-2,4,6-triene

Sol.



65. Correct statements regarding alkyl halides (R–X) among the following are :
(A) Alcohol being less polar solvent as compared to water, alcoholic KOH favours elimination reaction with R–X.
(B) Order of reactivity towards S_N1 mechanism is $C_6H_5–CH_2–Cl > C_6H_5–CHCl–C_6H_5$.
(C) Non substituted aryl halides exhibit properties similar to alkyl halides.
(D) Vinyl chloride is an example of haloalkene and allyl chloride is an example of haloalkyne.
(E) R–Cl can be prepared by reacting R–OH with $SOCl_2$ but Ar–Cl cannot be prepared by reacting Ar–OH with $SOCl_2$.
Choose the **correct** answer from the options given below :
(1) A, B and C Only
(2) B and D Only
(3) A and E Only
(4) D and E Only

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Sol. Ar-Cl and R–Cl shows different chemical properties

Rate of $S_N^1 \propto$ stability of first formed carbocation

Water is more polar than alcohol

Vinyl alcohol, allyl alcohol, both are alkene unsaturated alcohols

$R\text{–}OH \xrightarrow{SOCl_2} R\text{–}Cl$

Phenol does not give Ph–Cl with $SOCl_2$.

66. An organic compound "x" where molar ratio of C, O and H are equal, on treatment with 50% KOH under reflux followed by acidification produced "y". The most likely structure of "y" is:

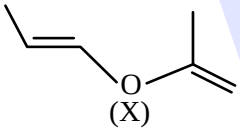
[Molar mass of 'x' is 58 g mol^{-1}]

(1) $CH_2{=}CH\text{–}\overset{\overset{O}{\|}}{C}\text{–}OH$ (2) $CH_3\text{–}CH{=}CH\text{–}CH{=}O$

(3) $O{=}\overset{}{C}\text{–}OH$ with $CH_2\text{–}OH$ (4) $CH_3\text{–}\overset{\overset{O}{\|}}{C}\text{–}OH$

Sol. $\begin{matrix}CHO\\|\\CHO\end{matrix} \xrightarrow[\text{Intramolecular cannizaro reaction}]{KOH} \begin{matrix}CH_2OH\\|\\COO^{\ominus}\end{matrix} \xrightarrow{H^{\oplus}} \begin{matrix}CH_2OH\\|\\COOH\end{matrix}$

67. A molecule (X) with following structure under mild acidic condition is hydrolysed to produce (Y) and (Z). Identify the correct statements about (Y) and (Z).



(X)

(A) Both (Y) and (Z) have same molar mass

(B) (Y) and (Z) can be distinguished from each other by $NaHCO_3$

(C) (Y) and (Z) react with HCN with same rates

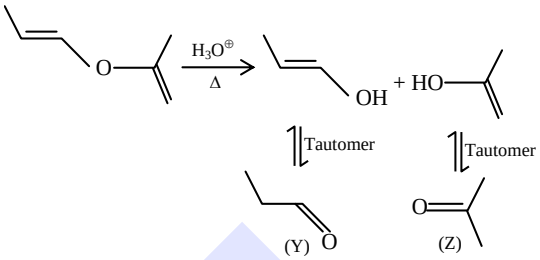
(D) (Y) and (Z) undergo addition reaction with 2, 4-DNP

Choose the **correct** answer from the options given below :

(1) A, B and C Only (2) B and C Only

(3) C and D Only (4) A and D Only

Sol.

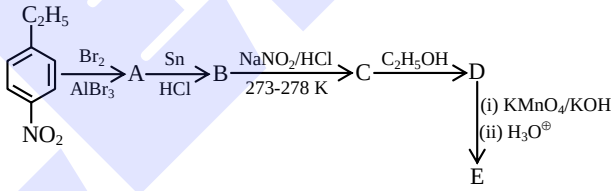


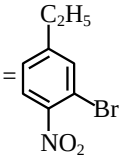
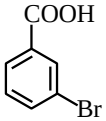
Y will give positive fehling's test

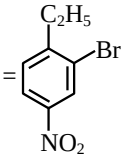
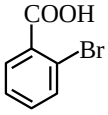
Molar mass of [Z] (C_3H_6O) = 58

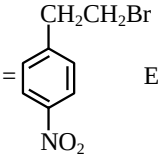
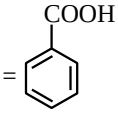
Molar mass of [Y] (C_3H_6O) = 58

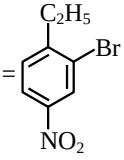
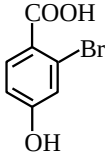
68. Identify compounds A and E in the following reaction sequence.



(1) A =  E = 

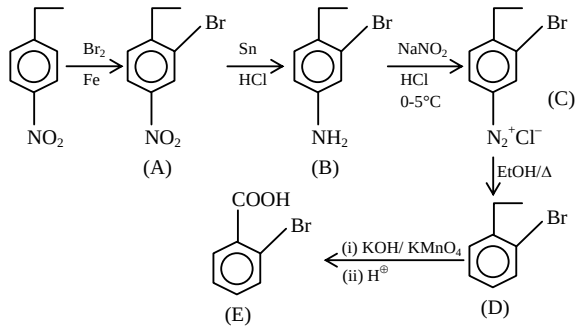
(2) A =  E = 

(3) A =  E = 

(4) A =  E = 

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Sol.



69. Identify the correct pair having amino acid (A) and the hormone (B) that is iodinated derivative of the amino acid (A).

(T and Y represent one letter code for amino acids)

Amino acid (A)	**Hormone (B)**
(1) T	Insuline
(2) T	Thyroxine
(3) Y	Thyroxine
(4) Y	Insuline

Sol. Thyroxine is derivative of Tyrosine and single letter symbol of Tyrosine amino acid is Y.

70. Among Fe^{2+}, Fe^{3+}, Cr^{2+} and Zn^{2+}, the ion that shows positive borax bead test and with highest ionisation enthalpy is:

(1) Fe^{2+} (2) Zn^{2+}

(3) Cr^{2+} (4) Fe^{3+}

Sol. Zn^{2+} does not show Borax Bead Test.

Among remaining ions; Fe^{3+} has maximum ionization energy.

SECTION-B

71. The surface of sodium metal is irradiated with radiation of wavelength x nm. The kinetic energy of ejected electrons is 2.8×10^{-20} J. The work function of sodium is 2.3 eV. The value of x is _____ $\times 10^2$ nm. (Nearest integer)

(Given : h = 6.6×10^{-34} J s; 1eV = $1.6\ 0 \times 10^{-19}$ J; c = 3.0×10^8 ms^{-1})

Sol. for photoelectric effect :

$$\frac{hc}{\lambda} = W + K.E.$$

$$\frac{6.6\times10^{-34}\times3\times10^{8}}{x} = 2.3\times1.6\times10^{-19} + 2.8\times10^{-20}$$

$$\frac{19.8\times10^{-26}}{x} = 3.96\times10^{-19}$$

$x = 5 \times 10^{-7}$

$x = 500 \times 10^{-9}$ m

$x = 5 \times 10^{2}$ nm

72. Consider the following gas phase reaction being carried out in a closed vessel at 25°C.

$2A(g) \rightarrow 4B(g) + C(g)$

time (min)	total pressure of the system (mm Hg)
30	300
∞	600

The pressure of C(g) at 30 minutes time interval would be _____mm Hg. (Nearest integer)

Sol.

$$2A(g) \rightarrow 4B(g) + C(g)$$

	p°		
t = 30 min	p°–2x	4x	x
t = ∞	–	2p°	p°/2

$p_\infty = 2p° + p°/2 = 600$

$5p°/2 = 600$

$P° = 240$ mm Hg

At t = 30 min, $p = 300 = p° - 2x + 4x + x$

$300 = 240 + 3x$

$3x = 60$

$x = 20$

Pressure of C(g) = 20 mm Hg

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73. Consider the following two half-cell reactions along with the standard reduction potential given :

$CO_2 + 6H^+ + 6e^- \rightarrow CH_3OH + H_2O \quad E^o_{red} = 0.02V$

$\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O \quad E^o_{red} = 1.23V$

The fuel cell was set up using the above two reactions such that the cell operates under the standard condition of 1 bar pressure and 298 K temperature. The fuel cell works with 80% efficiency. If the work derived from the cell using 1 mol of CH_3OH is used to compress an ideal gas isothermally against a constant pressure of 1kPa, then the change in the volume of the gas, $\Delta V =$ _____ m^3. (Nearest integer)

Given : F = 96500 C mol^{-1}

Sol. $E^o_{cell} = [1.23 - 0.02] = 1.21V$

$\Delta G° = -6 \times 96500 \times 1.21$ J

$W = \frac{80}{100} \times \Delta G° = -P\Delta V$

$0.8 \times 6 \times 96500 \times 1.21 = 1 \times 10^3 (\Delta V)$

$\Delta V = 560.472\ m^3$

74. Number of paramagnetic ions among the following d- and f-block metal ions is _____.

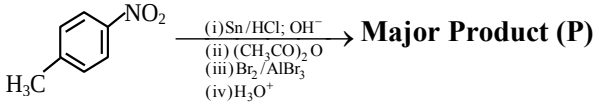
Mn^{2+}, Cu^{2+}, Zn^{2+}, Yb^{2+}, Sc^{3+}, La^{3+}, Gd^{3+}, Lu^{3+}, Ti^{4+}, Ce^{4+}

(Atomic number of Mn = 25, Cu = 29, Zn = 30, Yb = 70, Sc = 21, La = 57, Gd = 64, Lu = 71, Ti = 22, Ce = 58)

Sol. Paramagnetic : Mn^{+2}, Cu^{+2}, Gd^{+3}

Diamagnetic : Zn^{+2}, Yb^{+2}, Sc^{+3}, La^{+3}, Lu^{+3}, Ti^{+4}, Ce^{+4}

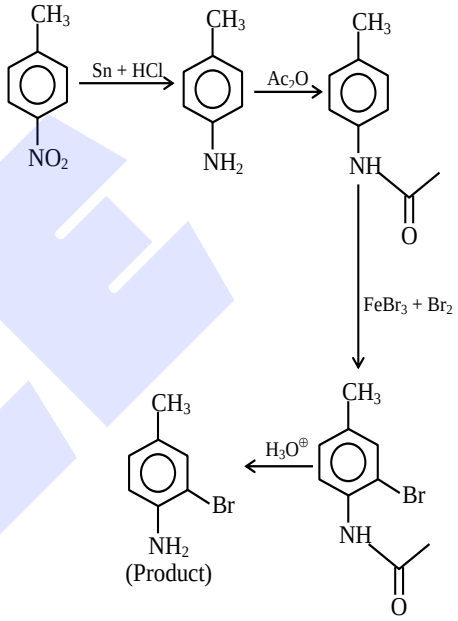
75. Consider the following reactions sequence



When the product (P) is subjected to Carius analysis using $AgNO_3$, 1.0 g of the product (P) will produce _____ g of the precipitate of AgBr. (Nearest interger)

(Given : molar mass in g mol^{-1} C : 12, H : 1, O : 16, N : 14, Br : 80, Ag : 108)

Sol.



Molar mass of product = 186

Mass of AgBr = $\frac{1}{186} \times 188 \approx 1$

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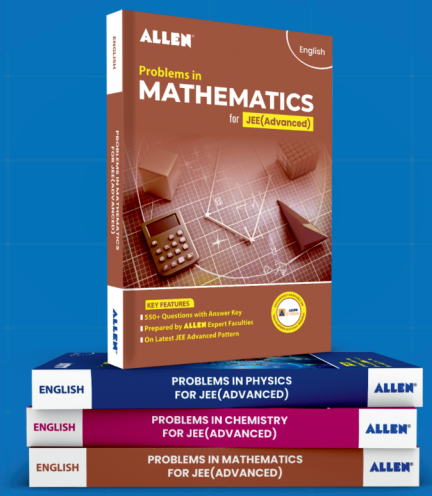


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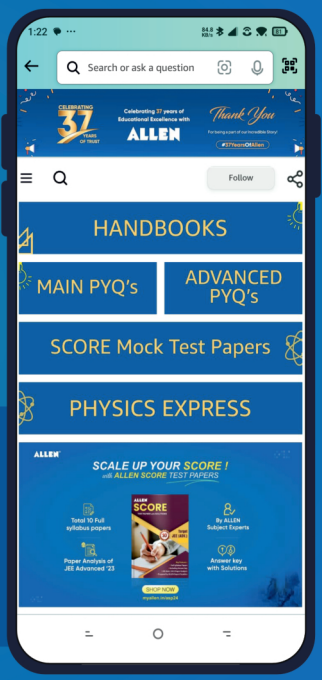
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